

Coding Wagara: Fostering Motivation through Generating Japanese Patterns using SF Symbols

Shinya Nishide

Senri International School of Kwansei Gakuin

Minoh-Shi, Japan

shinya.nishide@icloud.com

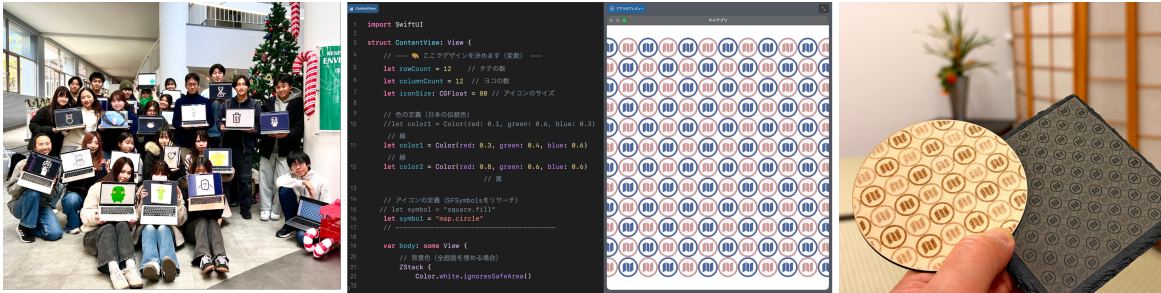


Figure 1: The assignment workflow showing CS1 high school students (left), coding Wagara patterns using SF Symbols in SwiftUI (center), and fabricating tangible laser-cut coasters (right).

Abstract

Transitioning from visual block-based coding to syntax-based programming is a critical challenge in CS1, where the gap between visual intuition and syntactic rigor often diminishes learners' confidence among novice middle and high school students. This assignment introduces an "Art-Thinking" framework grounded in Self-Determination Theory (SDT) that overcomes this barrier by removing the prerequisite for artistic drawing skills. By leveraging SF Symbols, a system-integrated library of over 6,000 vector icons, students perform procedural generation of Wagara (traditional Japanese patterns), transforming the abstract learning of loops and coordinates into a culturally responsive design process. The assignment culminates in dual tangible outputs: a laser-cut coaster (via Adobe Illustrator/SVG) and an iOS Visual Encyclopedia App. This approach successfully converts syntax anxiety into "productive struggle," empowering students to master complex logic through the intrinsic joy of creating high-fidelity, tangible art.

CCS Concepts

• Social and professional topics; • Computing education; K-12 education;; • Applied computing; • Arts and humanities; Media arts;; • Computing methodologies; • Computer graphics; Graphics systems and interfaces.;

Keywords

CS1, SwiftUI, Art Thinking, SF Symbols, Digital Fabrication, Wagara



This work is licensed under a Creative Commons Attribution 4.0 International License. *SIGGRAPH Educator's Forum '26*, Los Angeles, CA, USA
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2552-4/2026/07
<https://doi.org/10.1145/3799829.3812517>

ACM Reference Format:

Shinya Nishide. 2026. Coding Wagara: Fostering Motivation through Generating Japanese Patterns using SF Symbols. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH Educator's Forum '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3799829.3812517>

1 Introduction

Transitioning from block-based (e.g., Scratch) to text-based coding often diminishes learner confidence due to the gap between visual intuition and syntactic rigor [Weintrop and Wilensky, 2017]. While gamified environments provide initial engagement, research indicates this extrinsic motivation is often ephemeral due to the "novelty effect" [Hamari et al., 2014; Hanus and Fox, 2015]. Consequently, a critical pedagogical challenge is facilitating the internalization of motivation by shifting focus to intrinsic interest, as grounded in Self-Determination Theory (SDT) [Ryan and Deci, 2000].

To address this, this assignment introduces an "Art-Thinking" approach using SF Symbols—Apple's extensive vector icon library seamlessly integrated with SwiftUI. By leveraging over 6,000 icons as "digital paint," students utilize code for self-expression as a progression from puzzle-solving (Figure 1). This approach eliminates the need for traditional artistic skills and transforms syntax anxiety into the "productive struggle" of creative design.

2 THE ASSIGNMENT

The assignment spans 8–10 contact hours, structured to progress from static composition to dynamic app development. Key metadata is summarized in Table 1.

Phase 1: Symbol Composition (Art-Thinking)

Students navigate the SF Symbols library to discover shapes that inspire them. Using vector commands (`Image(systemName: "...")`), they combine symbols—for instance, utilizing a 'cloud' and

Table 1: Assignment Metadata

Summary	Students create algorithmic art using SF Symbols, process image data via Adobe Illustrator for fabrication, and develop an iOS Visual Encyclopedia App.
Learning Outcomes	Apply (control flow), Create (generative patterns), Develop (iOS App), Evaluate (aesthetic composition).
Classification	Fundamentals, Modeling (2D), Rendering (Vector), Applied Computing (Arts and Humanities).
Audience	Middle and High School Students (Novice / CS1).
Dependencies	Mac or iPad (Swift Playgrounds, Xcode), Adobe Illustrator, Laser Cutter (xTool etc).
Prerequisites	None. Designed for absolute beginners or students transitioning from block-based environments.
Strengths	Cultural relevance, immediate visual feedback, seamless transition to app development, high engagement through symbol exploration.
Weaknesses	Difficulty interpreting error messages, environment setup friction (permissions), steep learning curve for advanced app logic.
Variants	Vinyl stickers (Vinyl Cutter), T-shirt design (Heat Transfer), Greeting Cards (Print or Digital)
Assessment	Rubric-based: Code Complexity (Nested Loops), Visual Aesthetics (Color/Composition), Reflection (Design Intent).

'bolt' to construct a character. This enables even students without prior drawing skills to generate high-quality graphics, motivating them to write further code to realize their vision.

Phase 2: Algorithmic *Wagara* & Traditional Colors

Students transition to algorithmic thinking by recreating traditional Japanese patterns (e.g., *Ichimatsu* check patterns). They employ `VStack`, `HStack`, and nested `ForEach` loops to arrange symbols. Instead of standard colors, students define custom RGB values to replicate traditional Japanese colors, such as *Moegi-iro* (sprout green) or *Ebicha-iro* (maroon). Modifying a single loop parameter instantly transforms the visual impression, providing immediate feedback on abstract concepts.

Phase 3: Dual Output (Physical & Digital)

This phase connects code to the real world through two distinct outputs.

Physical (Coaster via Digital Fabrication): Students export their generated patterns as image data and import them into Adobe Illustrator. They learn to adjust the design into a 90mm diameter circle and process it for laser cutting (converting to SVG format). Finally, they use *xTool Studio* to fabricate wooden tea coasters. This process fosters a profound sense of agency as designers, as students observe their digital code materialized into a tangible product.

Digital (Visual Encyclopedia App): Students build a "Japanese Family Crest Encyclopedia" using Xcode. They list their generated patterns in a scrollable List view and implement navigation. Completing a functional app motivates them to self-scaffold advanced features such as Toggles and Buttons.

3 OBSERVATIONS AND REFLECTION

This assignment was implemented with 24 high school students (predominantly beginners).

Pre-class surveys indicated a tendency toward resistance, with 37.5% (9/24) of students explicitly mentioning "fear" or "worry" regarding text-based coding. However, mid-term reflections revealed a substantial attitudinal shift. The mean scores for "Fun/Play-like" ($M=4.38/5.0$) and "Sense of Drawing" ($M=4.17/5.0$) suggest that SF Symbols successfully bridged the gap between code and art.

Qualitative data observed a transition from "I'm scared" to "It's difficult, but fun," demonstrating productive struggle. Unlike traditional syntax drills, students voluntarily iterated on nested loops and coordinate parameters to refine their aesthetic outputs. The familiar context of SF Symbols acted as a scaffold, allowing students to persist through syntax errors to achieve their envisioned *Wagara* designs, effectively encouraging efforts to master CS1 concepts through an enjoyable creative process.

4 CONCLUSION AND FUTURE WORK

The results of this assignment suggest that integrating Art-Thinking with SF Symbols has the potential to effectively mitigate the "confidence drop" in CS1. By shifting the focus from puzzle-solving to the procedural generation of culturally relevant *Wagara*, students viewed code as a creative medium, with dual tangible outputs (coasters/apps) fostering intrinsic motivation.

However, limitations exist. The findings are based on a single implementation with 24 students. Additionally, novice learners still face discouragement when confronting cryptic build error messages, which occasionally hinders the "flow" state despite the visual nature of the symbols.

Future work involves replicating this curriculum in diverse settings (e.g., middle schools) and expanding fabrication options to vinyl stickers and T-shirt designs to broaden creative opportunities in computing education.

References

- Juho Hamari, Jonna Koivisto, and Harri Sarsa. 2014. Does gamification work? – A literature review of empirical studies on gamification. In *Proceedings of the 47th Hawaii International Conference on System Sciences (HICSS '14)*. IEEE Computer Society, 3025–3034, DOI: 10.1109/HICSS.2014.377.
- Michael D. Hanus and Jesse Fox. 2015. Assessing the effects of gamification in the classroom: A longitudinal study on intrinsic motivation, social comparison, satisfaction, effort, and academic performance. *Comput. Educ.* 80 (Jan. 2015), 152–161, DOI: 10.1016/j.compedu.2014.08.019.
- Richard M. Ryan and Edward L. Deci. 2000. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *Am. Psychol.* 55, 1 (2000), 68–78, DOI: 10.1037/0003-066X.55.1.68.
- David Weintrop and Uri Wilensky. 2017. Comparing block-based and text-based programming in high school computer science classrooms. *ACM Trans. Comput. Educ.* 18, 1, Article 3 (March 2018), 25 pages, DOI: 10.1145/3089799